

1. EXTENSIONS AND RELATED WORK

Throughout this section, fix positive integers m and n . By convention, $\sum_{\ell} v(\ell)$ always designates the sum of $v(\ell)$ over all (m, n) framed lattices ℓ . In all cases, v is a cell product function, and therefore Theorem ?? tells us that matrices can be used to evaluate such sums.

Fewer tiles. If we limit ourselves to a subset of the tiles in Figure ?? then, of course, there are fewer polygon mosaics and polygon free mosaics. We can count how many using our existing work. For example, omitting the last tile of Figure ??, the number of polygon mosaics equals $\sum_{\ell} v(\ell)$, where v is the cell product function that sends $\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}$, $\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}$, $\begin{smallmatrix} 0 & 1 \\ 0 & 1 \end{smallmatrix}$, and $\begin{smallmatrix} 1 & 0 \\ 1 & 0 \end{smallmatrix}$ to 0 and the other twelve cells to 1. This can be shown using the reasoning of Theorem ??, since the omitted tile is the one that maps via $\mathcal{L}$ to either $\begin{smallmatrix} 0 & 1 \\ 0 & 1 \end{smallmatrix}$ or $\begin{smallmatrix} 1 & 0 \\ 1 & 0 \end{smallmatrix}$.

We can similarly count the number of polygon mosaics that can be formed by omitting a different tile or by omitting several tiles. It is worth noting, however, that some of these turn out to be trivial. Specifically, if the second, third, fourth, or fifth tile is omitted from Figure ?? then there is only one possible polygon mosaic, namely the one that contains only blank tiles. Let us indicate how to see this in three ways for omitting the second tile; omitting the third, fourth, or fifth tile is similar. First, consider the tiles that are not blank, assuming there are some, in a polygon mosaic. Of those tiles, look at those in the highest possible row. The one farthest to the right must be the second tile of Figure ??. Second, in any framed binary lattice in which not all vertices are zero, among the vertices that are 1 and are as high as possible, the one farthest to the right must be the corner of a cell that is $\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}$. Finally, in Theorem ??, if $v\left(\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}\right) = 0$ and $v\left(\begin{smallmatrix} 0 & 0 \\ 0 & 0 \end{smallmatrix}\right) = 1$ then for all positive integers m and n , the upper right entry of A_n^m equals 1.

As for determining the number of polygon free (m, n) mosaics using a reduced tile set, this will simply be 7^{mn} if the second, third, fourth, or fifth tile is omitted from Figure ??, since we have just seen that no polygons can be formed. For these or other omitted tiles, we can use the ideas from Theorem ??. For example, if we omit the last tile of Figure ?? then we use the cell product function v that maps $\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}$, $\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}$, $\begin{smallmatrix} 0 & 1 \\ 0 & 1 \end{smallmatrix}$, and $\begin{smallmatrix} 1 & 0 \\ 1 & 0 \end{smallmatrix}$ to 0, $\begin{smallmatrix} 0 & 0 \\ 0 & 0 \end{smallmatrix}$ and $\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}$ to 6 (since there are six tiles that can replace the blank tile in a polygon free mosaic), $\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}$ and $\begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}$ to -1 , and the remaining eight tiles to 1. For omitting the blank tile, we would use v defined by mapping $\begin{smallmatrix} 0 & 1 \\ 0 & 1 \end{smallmatrix}$ and $\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}$ to 0, $\begin{smallmatrix} 0 & 0 \\ 0 & 0 \end{smallmatrix}$, and $\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}$ to 6, $\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}$ and $\begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}$ to -1 , and the remaining ten tiles to 1. In this case we would consider polygon mosaics that might include blank tiles, and create only those polygon free mosaics in which the blank tile was replaced by a tile that is not blank. This example was posed by the first author of this paper and served as the motivation for all our work.

Adding tile variations. Consider the variations of the final tile of Figure ?? that are in Figure [to be created—all teeth face the same direction, unlike in the current figure at the end of the paper, but we picture one and two “teeth”], where we add one, two, or any number of “teeth.” If we add $a - 1$ such tiles, so that there are a tiles that connect the midpoints of the top and bottom tile edges, then the number of polygon mosaics then becomes $\sum_{\ell} v(\ell)$, where v is defined by mapping $\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}$ and $\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}$ to 0, $\begin{smallmatrix} 0 & 1 \\ 0 & 1 \end{smallmatrix}$ and $\begin{smallmatrix} 1 & 0 \\ 1 & 0 \end{smallmatrix}$ to a , and the remaining twelve tiles to 1. Similarly, the number of polygon free mosaics is $\sum_{\ell} v(\ell)$, where v is defined by mapping $\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}$ and $\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}$ to 0, $\begin{smallmatrix} 0 & 1 \\ 0 & 1 \end{smallmatrix}$ and $\begin{smallmatrix} 1 & 0 \\ 1 & 0 \end{smallmatrix}$ to a , $\begin{smallmatrix} 0 & 0 \\ 0 & 0 \end{smallmatrix}$ and $\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}$ to $a + 6$, $\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}$ and $\begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}$ to -1 , and the

remaining eight tiles to 1. To see both of these results, extend the function $\mathcal{L}$ to act on this expanded set of mosaics. Fix an (m, n) framed lattice ℓ . When counting polygon mosaics, $v(\ell)$ is the number of polygon mosaics that map to a given framed binary lattice ℓ , whereas when counting polygon free mosaics, $|v(\ell)|$ is the number of polygon mosaics mapping to ℓ , and the sign of $v(\ell)$ is as in Proposition ??.

Adding other tiles would work similarly, provided we are careful with the signs when counting polygon free mosaics. For example, if we took the expanded tile set above and additionally added $b - 1$ variations of the second tile of Figure ??, for a total of b such tiles, then the number of polygon free mosaics would be $\sum_{\ell} v(\ell)$, where v is the cell product function mapping $\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}$ and $\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}$ to 0, $\begin{smallmatrix} 0 & 1 \\ 0 & 1 \end{smallmatrix}$ and $\begin{smallmatrix} 1 & 0 \\ 1 & 0 \end{smallmatrix}$ to a , $\begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}$ to -1 , $\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}$ to $-b$, $\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}$ to b , $\begin{smallmatrix} 0 & 0 \\ 0 & 0 \end{smallmatrix}$ and $\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}$ to $a + b + 5$, and the remaining seven tiles to 1.

Tiles with more than one line. Consider adding one or more of the following four tiles to the tile set of Figure ??: [four tiles; first tile has connections from left edge to upper edge and also from right edge to lower edge; second tile has connections from left edge to lower edge and from right edge to upper edge; third and fourth tiles are the two crossing tiles that make up knot mosaics]. Using these tiles to form a mosaic gives us additional possibilities for polygon mosaics. Note that using the third or fourth tile allow us to form polygons that overlap, but each segment is still part of a unique polygon. We can still define $\mathcal{L}$ as we did in Section ?? by counting the number of polygons surrounding a vertex, and these new tiles all map to either $\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}$ or $\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}$.

If we add one of these tiles then the number of polygon mosaics is $\sum_{\ell} v(\ell)$, where v is the cell product function that maps all cells to 1, so that $\sum_{\ell} v(\ell) = 2^{(m-1)(n-1)}$. If we add two, three, or four of these tiles then the number of polygon mosaics is $\sum_{\ell} v(\ell)$, where v is the cell product function that maps $\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}$ and $\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}$ to 2, 3, or 4, respectively, and all other cells to 1.

If we specifically add the last two of these four tiles, then the question of counting polygon mosaics is exactly the problem investigated by Oh et al. [Add citation and more historical context here.]

Although we could consider polygon free mosaics adding with tiles in Figure [the one above including knot tiles], we will not do so. We invite others to find a way to count such mosaics, but we indicate some potential difficulties. First, there are two reasonable ways to define a polygon free mosaic. A polygon free mosaic could either be a mosaic in which no polygon can be found, or one in which no set of tiles exists for which all segments in those tiles are part of polygons. For example, the following mosaic is polygon free in the second sense but not in the first. [A picture needs to be created: a $(2, 2)$ mosaic that has a diamond polygon, but one of the tiles has an extra diagonal segment.] A second problem is that the end of the proof of Proposition ?? relies on polynomials not being able to cross one another, something that is possible if either the third or fourth tiles in Fig [new one with knot tiles] are used.

Shaded tiles. Consider a tile set containing fourteen tiles. In addition to the blank tile, there would be an all-shaded tile, and then for all non-blank tiles in Figure ??, we would substitute two tiles, with shading on one side of the line or the other. For example, the final tile in Figure ?? would be replaced by [new diagram; left tile has shading on right, right tile has shading on left]. We would define a

polygon mosaic to be one in which both the line segments and the shading matches on adjacent tiles, and with an entirely unshaded outer boundary. The function $\mathcal{L}$ would be defined exactly as in Section ??, i.e., ignoring all shading. The result is that all vertices in unshaded regions are 0 and vertices in shaded regions are 1. The number of polygon mosaics would be exactly what is in Theorem ?. While this is uninteresting in itself, the shading allows us to make variations by adding or subtracting tiles. For example, if we omitted the left shaded tile above then the number of polygon mosaics would be as in Theorem ?, except that we would have v defined with $v \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix} = 0$. Shaded tiles can also be used in various counting problems involving polygon free mosaics.